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Chapter Author(s): Laura Vaughan

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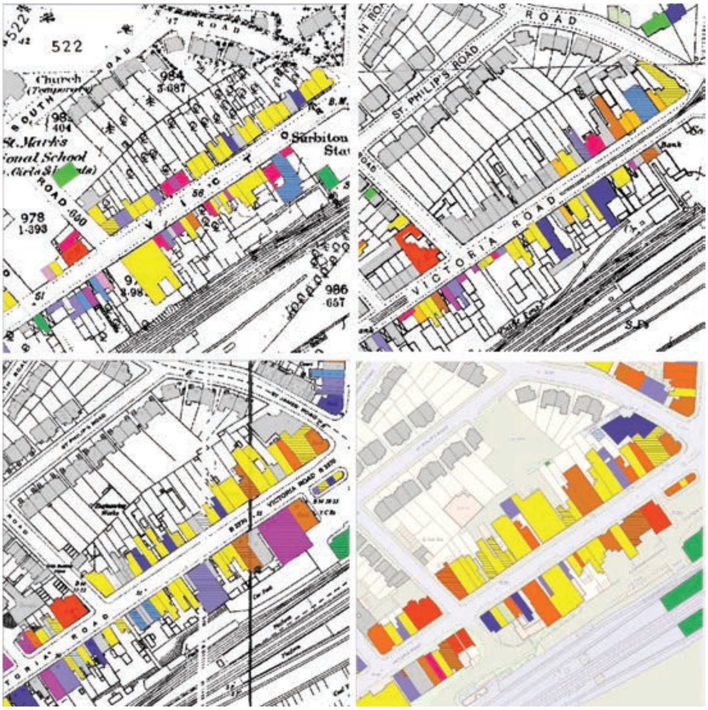
PART C
High Street Diversity



Non-domestic land uses, Victoria Road, Surbiton

Chapter 7

High Street Diversity

Laura Vaughan

I. Introduction

The nursery rhyme of the butcher, the baker and the candlestick-maker is increasingly used nowadays to emphasise the importance of diversity in the high street. Broadly equivalent to the main street in the USA, the high street is a common metonym used in public debate when discussing town centres and is frequently alluded to when discussing the apparent decline of smaller centres. As shown by Griffiths in his chapter for this volume, discussions of town centre decline typically manage to convey two messages at once: that the traditional town centre contains a variety of uses, and that those uses will include both shops and small-scale craftsmen or manufacturers. This is frequently set in contrast to cases where high streets are transforming into bland 'clone towns' comprised of large chains or where high streets are in decline, comprised of charity shops and shuttered business units. Or, alternatively, it is presented in contrast to the supposed move of retail online, with the 'death' of the high street approaching fast. Elsewhere in the world, particularly in North America, the death of suburbia's centres has apparently already occurred, despite strong evidence to the contrary from research conducted by, e.g. Keil and Shields (2013), and indeed the infinite variety of suburban life continues to spread world-wide.

One solution to the decline of some suburban centres is diversity. It is a measure of success increasingly cited in government reports and academic research alike, and earlier research has suggested how multi-functional town centres sustain adaptability over time: with evidence that centres are much more resilient if they are not reliant on just one form of activity (Vaughan et al. 2013). But the reality is that diversity is rarely measured with any great accuracy or detail and measures of diversity that take account of a spatial dimension are rarer still. After all, diversity has innumerable spatial variations. If we take land uses as an example, diversity can take the form of mixing at the scale of a block face, two sides of a street, around a corner or within a certain distance. Each of these has consequential social and economic relationships, so a mix of uses across a town centre does not necessarily mean that uses are mixed along a street alignment. The challenges of measuring diversity and its significance for suburban studies lie at the heart of the research described here.

In her influential book *The Death and Life of Great American Cities*, Jane Jacobs (1961) proposed four conditions for creating city diversity: a mixture of primary uses within each district and within each part of the district; short blocks and multiple intersections; a variety of buildings of different types and ages; and a dense concentration of people, including people resident in the district. Her ideas recur in every decade since – probably as they are such a powerful argument against mono-functional districts. For suburban settings such as those described in this book, Jacobs' description of the daily ballet of people as doing little more than coming and going is even more important, as it demonstrates how public life is comprised of the prosaic, small-scale activities that collectively make a trip into the local centre worthwhile. An understanding of the 'rhythmic landscape' of cities, comprised of differing intensities of activity, and different uses and people, has been taken up by more recent writers, including John Allen (see Massey et al. 1999), who emphasises how even changing patterns of light and smells recompose the city hour by hour. More recent research by Palaiologou and Vaughan (2012) on the evolution of Manhattan's row houses and Islington's terraced houses provides empirical evidence on how spatial boundaries between house and street and the street's own urban setting in turn, create shifting patterns where the individual is (successively or abruptly) transformed from inhabitant, to commuter and to citizen as they move from house, to local street, and to the city at large. The latter research shows how diversity of uses contributes to this shifting pattern, by bringing local and stranger into contact with one another. The diversity of uses creates a rhythm, where, as one walks down a street, a regular change in the *visual landscape* creates a natural setting for an intermingling of people on their daily perambulations.

It is interesting to continue the language of landscape and to consider an analogy between hedgerows and town centres. Ecological research proposes that hedgerows tend to have a much greater diversity of plants and animals and to be thicker, taller and more continuous as they increase in age (Hooper 1970). This chapter proposes that there is an analogous consideration of diversity and age that can be made by conceiving suburban high streets as 'hedgerows'. Beyond the rather neat association between leafy suburbs and green hedges, such a conception implies the emergence of distinctive, complex and stubbornly persistent material cultures over time. As Alan Penn and colleagues have pointed out, diverse areas are not formed randomly, but are the outcome of highly structured sets of related systems (social, economic and cultural) as well as physical (Penn et al. 2009). Also referring to ecological theory, they state how the outcome of the long-term formation of diversity is not a random mixing of uses, cultures and economies, but a distinct pattern that follows a spatial logic. Just as in ecological systems theory, where the richness and evenness of species in a community (i.e. the number of different types of species and the number of each species in the same area) are said to contribute to the overall resilience of the ecosystem, the following historical morphological analysis, together with an exegesis of historical business directories, explores the extent to which the diversity of suburban town centres is associated with their historical pattern of land use and habitation.

In one of the most famous studies in the field of urban morphology, Conzen (1960) showed how the mediaeval burgage plots of the town of Alnwick went through a cycle of continuity and change over time, with shifting patterns of plot sub-division into open-plan, multiple use, rear courtyard uses and partial consolidation. This is not a mere function of built form or urban tissue. The underlying land pattern that follows this pattern of continuity and change has its own life-cycle, but importantly, this life-cycle maintains certain generic spatial relationships. As was proposed so lucidly by the late architect Richard MacCormac in his *Anatomy of London* (1996):

> there seems to be a pattern in the relationships which reoccurs over time though the functions change. For example, in the eighteenth-century-city large houses on primary streets were inhabited by high-income families and the mews behind serviced them. Today the houses might be offices with the mews inhabited by businesses selling services – commercial or professional – like photocopying, printing or sandwich bars to the primary users.

According to MacCormac, such streets are 'like reinhabited coral reefs' (MacCormac 1996: 307). This built form, land use and spatial configurational interdependence are not trivial matters, since just as much as coral reefs constitute delicate ecologies that can weather minor changes in sea currents, predators and the like (and the reader can easily discern a similarly effective analogy to be made with regard to hedgerows), a major disruption to that ecology can be life-threatening. Understanding the nature of diverse ecology in the context of the suburban high street is considered in this chapter.

Another theoretical approach needs explaining at this juncture: the way in which the built environment is considered as a configurational network, one in which, rather than considering buildings, plots or streets in isolation, the relationship of a single street to its surroundings is measured using space syntax analysis. In this way, diversity is considered as a spatially related factor, rather than a random mixing of land uses (or people) within an area. As mentioned above, diversity can take different forms according to its spatial context (see also Marcus and Ståhle 2005; Sevtsuk and Mekonnen 2012). If one considers the differences in experience between, say, walking down a street with five different uses on both sides of the street, to that of a street where only a single use is observed on either side, but four others can be found round the corner, the importance of bringing the spatial detail into what would otherwise be dismissed as 'mixed-use' becomes evident. This is even more the case when one considers the way in which different uses set up different potential connections between building and street and the people using that street.

This chapter examines four periods of London's development to provide evidence for a new proposition on how high streets adapt to change.[1] It opens with a discussion of the need to distinguish between mixed-use in general and spatial diversity in particular: the fact that diversity has many dimensions that arguably work together to make for a rich ecology of the high street experience. Analysis of shifting diversity patterns over time is shown to correspond to the continuity and robustness of the street network. Seeing architecture as an object – where the mixing of different people carrying out different activities makes for the rich experience of encounter and daily routine – is shown to be a mundane but vital aspect of what makes high streets places that continue to be valued. A focus on building diversity ends the chapter, illustrating how the persistence of street form is tied up with how buildings and plots are arrayed along the high street.

II. Concepts of diversity

Notions of diversity cannot be separated from the phenomenological properties of urban and suburban built environments. These are consistent with claims made by Hillier et al. (1987) that the fundamental intelligibility of the built environment is an essential condition of bringing people together in public space and affording them encounter. In space syntax theory 'intelligibility' refers to the ability of certain configurations to allow for an overlap between people carrying out local activities on foot, alongside people passing through on their way to longer-distance trips. Other research by Penn et al. (2009) – building on Hillier's proposition regarding centrality being part of a spatial–temporal process (Hillier 1999a) – has shown for example that areas of the grid with a greater amount of land use diversity have spatial properties that allow for greater amounts of circulation, with a denser and more intensified local grid (Zhang 2005).[2] Penn et al. point to another essential aspect of diversity: this is not purely an aspect of land use distribution, and in order to have land use diversity, they argue, an urban area must 'support a diverse social and economic life as well, with all the outward manifestations of that which are open to the sensory experiences of sound, sight and smell' (Penn et al. 2009: 220). They suggest that experiential diversity is formed by differentiating building shapes, land parcels and movement patterns, and consequently shaping the pattern of human encounters. The importance of the experiential dimension of the built environment, of 'human bodies in material space', has also been highlighted by the phenomenologist David Seamon, who maintains that the vibrancy of urban places is supported or stymied by the material or spatial qualities such as street configuration, population density and mix of activities (Seamon 2013: 143). This rich tapestry of the architectural–social morphology of town centres, alongside the importance of seeing such places as emergent – that is, as having developed their spatial/experiential qualities over time (as pointed out by Griffiths elsewhere in this volume) – is the approach used here: intending to refute the apparently conventional nature of these sites and enquire into the forces that help shape their inherent complexity.

III. Land use diversity

This chapter describes research that took place as part of a four-year study of a sample of London's outer town centres. This built on a previous three-year study, which looked into the development of London's suburbs both spatially and economically to find the factors that make their town centres and surroundings successful and vibrant. The four cases were chosen from each of the outer London quadrants: north-west (High Barnet), north-east (Loughton), south-east (South Norwood) and south-west (Surbiton), and detailed data on historic and contemporary land uses, changes in street network configuration and built form were compiled into a large geographical information database to allow for detailed analysis of the factors that contributed to the centres' success. This was coupled with a contemporary ethnography of two of the cases.

The first project's research into London's historical growth found that its suburbs had a high degree of continuity in street networks over time, providing a relatively robust network. This can be seen on the sequence of maps of Surbiton's evolution since the 1820s (Figure 7.1). Whilst in the earliest period Surbiton was the name for the small junction of streets south of Kingston-upon-Thames, by 1880, though still a semi-rural area, it had loci

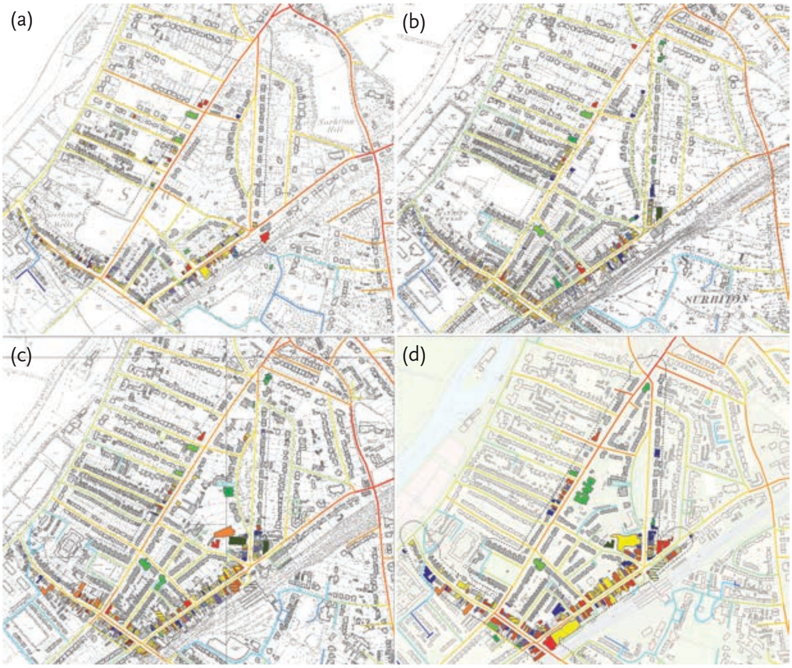


Figure 7.1
Non-domestic land uses in Surbiton (a) 1880, (b) 1910, (c) 1960 and (d) 2013, overlaid with segment angular integration at 800m. Refer to Fig. 7.4 for land use legend

of settlement that are recognisable today. The sequence of images shows intensification of the network, with particular building features (including garages) reshaping the network to accommodate new ways of living; it is only in the contemporary map that the planned interventions in the twentieth century are seen to have had a profound effect on the larger-scale connectivity of the area when the Kingston By-Pass enabled the creation of a focus of activities reliant on car travel and corresponding to wider patterns of accessibility, but not local ones at all, such as a traveller's hotel (see more on this in the chapter by Dhanani in this volume).

If we peel back to the past we can see how the availability of passing movement changes with the evolution of the street network; it in turn adapts its configuration to serve land uses of different types, according to the most beneficial catchment distances. Space syntax analysis considers the urban network as a spatial configuration, proposing a fundamental relationship between spatial morphology, movement patterns and the distribution of land uses as a 'movement economy'. The synergy of activities in and around suburban town centres has been shown to stem from the presence of overlapping movement flows of through-movement (represented by a statistical correspondence between the space syntax measures of choice and integration). This overlapping of different types of movement is proposed to create the conditions for local diversity in land use. So in Figure 7.1 we see that in 1800 the original settlement of Surbiton (on the north-western edge of the map) was connected to the larger-scale network through roads leading to the south-west (on the edge of the River Thames) and to the south-east. By the time the railway arrived in 1890, new connections had been created to serve a different set of scales of movement and activity.

In parallel to this shifting of scales, some sections of the network show features of a path dependency, with certain generic characteristics emerging, so that a particular trajectory of a business or industrial activity ends up persisting over time (Verburg et al. 2004; Page 2006; Griffiths et al. 2009). For example, Ewell Road in Surbiton served doctors and other such professional and commercial services in nineteenth-century Surbiton (*Kelly's London Suburban Directory* 1901)[3] and today it serves a very similar range of activities and services, although with more restaurants than in the past.

In the case of the research described here, the relationship between the structure of the built environment and functional land use distribution in suburban town centres was associated with its spatial setting by recording all non-domestic land use and constructing a dataset in which all the street sections were listed, and then attributing the spatial unit to the buildings lying alongside it (or vice versa). Diversity was then modelled in differing ways: considering the relative concentration of non-domestic uses in particular streets, taking account of frontage length or building footprint area as well as the degree of land use mix street by street across the town centre – either according to land use *group* (of which there are five) or by the finer-detailed *class* (of which there are theoretically twenty-six, although none of the cases had more than twelve in any of the studied periods).

The first analysis looks at the pure mixing of activities across a large area of each of the four cases. The areas were determined by drawing a notional line around all the streets which had continuous non-domestic activity from 1890 to 2013, so there might have been streets excluded from the analysis if their non-domestic uses did not exist in more recent periods. The distribution of the different classes of use over time was analysed by drawing bar charts of weighted counts of land uses in High Barnet, South Norwood and Surbiton (Figure 7.2). These graphs show an increase in the proportion of street segments with the maximum of four or five use classes in the latter two periods and a commensurate decrease in streets with only

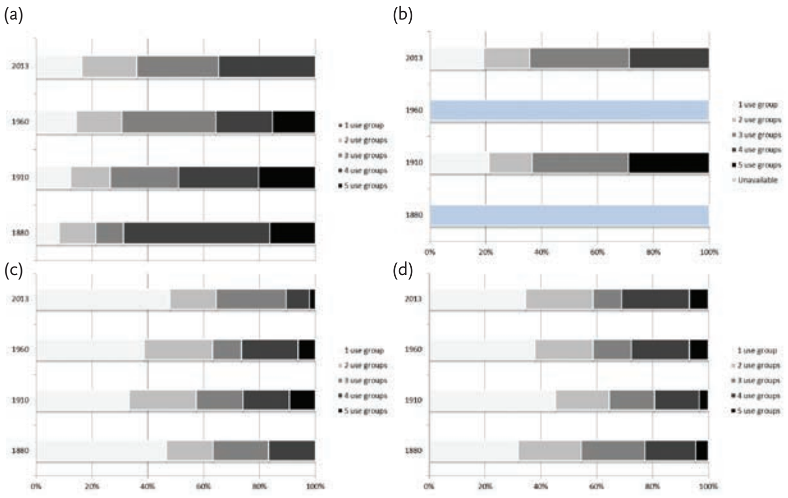


Figure 7.2
Weighted counts of land uses in the four study areas in (a) High Barnet, (b) Loughton, (c) South Norwood and (d) Surbiton across time
Laura Vaughan.

one use. Notably, both High Barnet and Surbiton – the two centres with greater economic success today (not only by the subjective assessment of their more vibrant appearance, but also in their possessing a larger town centre area according to national town centre statistics) – also have an increase in the proportion of streets with a mixing of uses over time. We might conclude from this that diversity of use goes hand-in-hand with economic success over time.

The location of these streets with diverse uses is also important when taking account of the experiential dimension of land use mix. Taking Surbiton, for example (Figure 7.3), it is clear that non-domestic land uses are spread out through the area right from the earliest period, but mixing of uses occurs principally on Victoria Road and Brighton Road, the two streets which form a V to the south of the images, whilst a small section of Maple Road, running north-east, has a focus of activity through the period, but with a lower level of mixing.

In Table 7.1 the streets with the greatest concentration of non-domestic land use activity are separated from those with fewer such uses (so these might be streets with a church and a school, with the remaining buildings being purely residential). Evidently there is a big difference in building area between the two groups. Whilst frontage length becomes more even over time, we see the emergence of larger buildings overall, but also a focus on certain uses, such as retail. Surbiton, which of all the cases has the most varied and constant diversity of activity all the way through to 2013 (Figure 7.4),

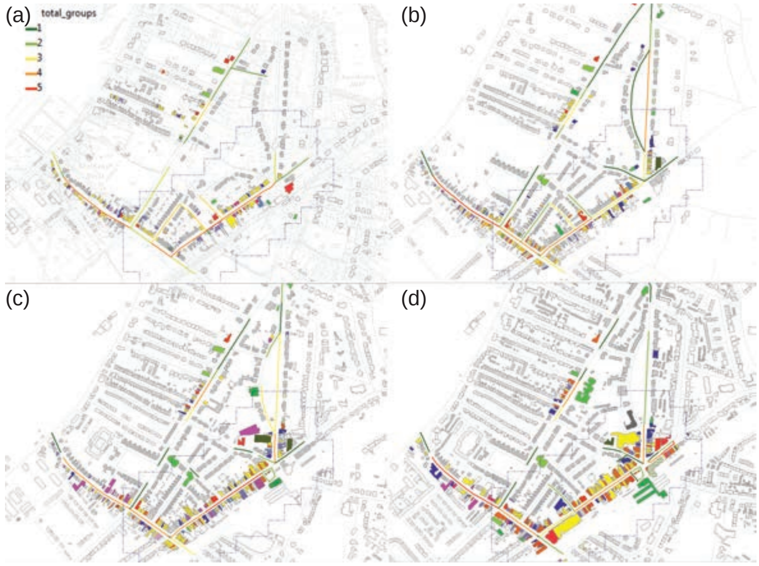


Figure 7.3
Non-domestic land uses around Surbiton town centre in (a) 1880, (b) 1910, (c) 1960 and (d) 2013, with road centrelines coloured to indicate the total number of land use groups in each street section. Refer to Fig. 7.4 for land use legend.
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also has the most consistency in the width of frontages, an indication of the amount of potential for diversity available in the building alignment. These differences are picked up in the penultimate section.

IV. Movement interfaces

Hillier et al. suggested in an early space syntax study that it is likely that a correlation between the mathematical values of spatial integration and choice might 'index the degree of correlation between these two types of movement pattern ... the degree of "movement interface" between inhabitants and strangers' (Hillier et al. 1987: 237). However, it is only in recent years that the analytic capability has been developed to examine this relationship. Whilst choice approximates a measure of through-movement potential and integration approximates a measure of to-movement potential, the locations where the highest values for both measures overlap to the greatest degree will create, as termed by Hanson, 'different modes of spatial co-presence and virtual community' (Hanson 2000: 115). It could be

Table 7.1 Non-domestic frontage length and building area

	Non-domestic frontage length (totals in metres)			Non-domestic building area (average for each group in turn m²)		
High Barnet	**Upper intervals**	**Lower intervals**	**All non-domestic**	**Upper intervals**	**Lower intervals**	**All non-domestic**
1880	744.17	506.11	1250.28	949.71	422.82	648.63
1910	819.07	1351.78	2170.85	1178.14	645.69	785.86
1960	1148.43	1466.27	2614.72	1157.87	1034.84	1079.85
2013	830.77	1227.01	2057.77	1499.69	726.9	939.41
Loughton	**Upper intervals**	**Lower intervals**	**All non-domestic**	**Upper intervals**	**Lower intervals**	**All non-domestic**
1880	*Data unavailable*	*Data unavailable*	*Data unavailable*	*Data unavailable*	*Data unavailable*	*Data unavailable*
1910	311.31	889.65	1200.95	663.56	552.3	575.48
1960	*Data unavailable*	*Data unavailable*	*Data unavailable*	*Data unavailable*	*Data unavailable*	*Data unavailable*
2013	1140.83	996.44	2137.27	2278.17	750.67	1181.5
South Norwood	**Upper intervals**	**Lower intervals**	**All non-domestic**	**Upper intervals**	**Lower intervals**	**All non-domestic**
1880	458.03	716.9	1174.93	789.48	281.23	348.68
1910	1350.96	1198.48	2549.44	1070.82	455.92	626.73
1960	1291.11	1242.76	2533.87	1286.56	510.68	716.53
2013	810.6	1879.9	2690.5	1028.13	864.87	902.28
Surbiton	**Upper intervals**	**Lower intervals**	**All non-domestic**	**Upper intervals**	**Lower intervals**	**All non-domestic**
1880	742.48	413.59	1156.07	1349.03	389.66	578.58
1910	825.43	597.96	1423.39	1452.16	325.36	579.8
1960	803.71	892.68	1696.39	2277.45	626.85	968.35
2013	950.75	1346.98	2297.73	2754.92	992.57	1357.19

said that integration is a measure of exploratory movement, whilst choice is highly influenced by the scale (or distance) at which you are measuring and it will tend to highlight major road networks – the deep structure of the network. Here we repeat this analysis in greater detail by considering where peak correlations exist for each case and across the four periods. This analysis of co-presence is carried out through time to ascertain the changing peak locations of co-presence in the network.

As has been proposed in an earlier piece using this form of analysis (Vaughan et al. 2010b), the location of greatest overlap is likely to be where the qualities of centrality associated with suburban movement economies are most likely to be seeded. We termed this the 'spatial signature' of suburban town centres, suggesting that some centres operated best for a much more

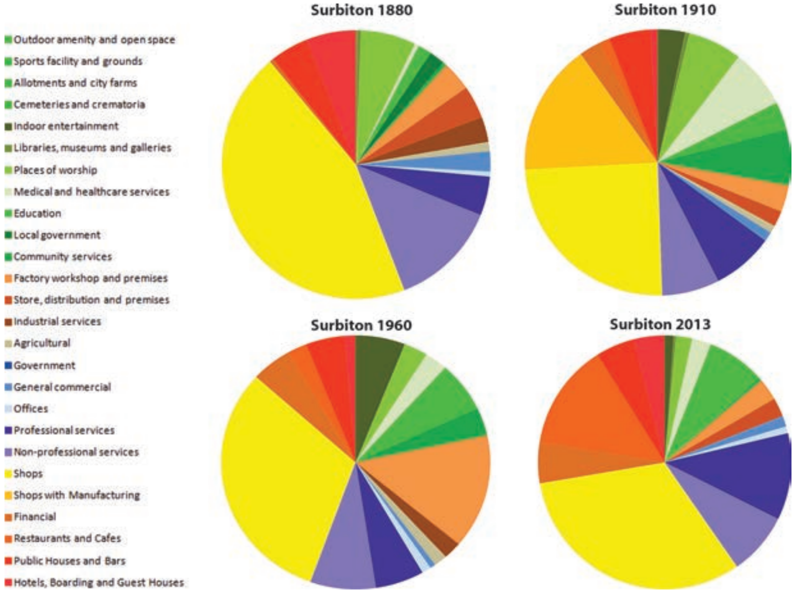

Figure 7.4
Pie charts showing average building area by use class for Surbiton in 1880, 1910, 1960 and 2013
Laura Vaughan.

localised network of connections, whilst others sustained connections both locally and more widely. Where the two correspond closely, two sorts of activities can easily coincide during the same trip: both moving from a particular origin to destination and en route doing other things, going different ways, depending on whatever combination of planned (and unplanned) activities is desired. In such a network, the same trip to the local supermarket can take a different route each time, depending on an individual's specific plans that week or indeed what the grid offers up opportunistically. A limited amount of redundancy means that pedestrian activity spreads out beyond a single street and, given the right conditions, enables interactions to flourish accordingly.

Bearing in mind that the following analysis focuses on the set of streets and their immediate environs that had continuous non-domestic inhabitation over the past 130 years, Table 7.2 shows the highest correlation between integration and choice at the same radius (e.g. choice 400 correlated with integration 400), selecting only the streets within the comparative boundary, highlighting the top three correlations for each epoch in bold. Two clear findings emerge: first, focusing on the figures in bold, with the exception of South Norwood, the cases have a distribution of high as well as low scales within which there is a correspondence throughout the periods (so for example, High Barnet has its top three correlations at scales of 3000 and 4000 as well as 800 or 1200 in any given period). If we then focus only on correlations above .750, which are at the higher end, only High Barnet and Surbiton have any such cases. It might be said that this is a feature of these two centres, which are the more successful. The meaning of this on the ground is that there are two scales – one local and one wider-reaching – at which the two networks of activity overlap. Less correspondence means that there are fewer opportunities for people carrying out varying journeys to intermingle – large-scale trips occur in parallel to people moving locally and the latter have less opportunity to explore beyond their immediate neighbourhood. Where they do intermingle with wider-scale trips, this occurs

Table 7.2 Highest correlation between integration and choice at the same radius, selecting only the streets within the comparative boundary, highlighting the top three correlations for each epoch in bold

High Barnet	**400**	**800**	**1200**	**1600**	**2000**	**3000**	**4000**
1880 (n=47)	0.610	**0.817**	**0.831**	0.724	0.741	**0.796**	0.750
1910 (n=57)	0.487	0.695	**0.751**	0.692	0.704	**0.840**	**0.820**
1960 (n=60)	0.318	0.643	0.616	0.641	**0.700**	**0.808**	**0.813**
2013 (n=68)	0.435	**0.732**	0.727	0.680	0.726	**0.837**	**0.846**
Loughton	**400**	**800**	**1200**	**1600**	**2000**	**3000**	**4000**
1880 (n=27)	0.434	0.436	0.577	**0.706**	**0.746**	**0.664**	0.686
1910 (n=36)	0.568	0.494	0.574	**0.681**	**0.748**	**0.683**	0.653
1960 (n=52)	0.548	0.709	0.639	**0.727**	**0.751**	**0.548**	0.482
2013 (n=69)	0.646	**0.716**	0.565	0.581	0.644	**0.715**	**0.705**
South Norwood	**400**	**800**	**1200**	**1600**	**2000**	**3000**	**4000**
1880 (n=58):	**0.695**	**0.651**	**0.631**	0.607	0.563	0.588	0.612
1910 (n=90):	0.627	**0.698**	**0.665**	**0.660**	0.622	0.535	0.609
1960 (n=94):	0.642	**0.727**	**0.723**	**0.697**	0.669	0.633	0.661
2013 (n=107):	0.639	**0.742**	**0.717**	**0.708**	0.696	0.633	0.600
Surbiton	**400**	**800**	**1200**	**1600**	**2000**	**3000**	**4000**
1880 (n=41):	0.386	0.761	0.668	**0.814**	0.736	**0.778**	**0.763**
1910 (n=48):	0.567	**0.776**	0.702	**0.858**	**0.813**	0.740	**0.790**
1960 (n=61):	0.417	**0.774**	0.653	0.750	0.728	**0.815**	**0.787**
2013 (n=76):	0.674	**0.892**	**0.862**	**0.777**	0.745	0.761	**0.777**

in streets that were not intended for this purpose, such as in the high street, South Norwood, where on a typical day pedestrians can be overwhelmed by the volume of heavy goods vehicles and buses making trips across the city, polluting the atmosphere for people walking down the narrow pavements to the local café, office, high school or shops (Figure 7.5). In the following, the way in which the suburban centres are embedded in major road networks (many of which are historical) is studied by plotting the above values in line charts which show the shift in degree of overlap over time for each centre. Figure 7.6 shows the result of averaging the R^2 value across the four periods.

We see that South Norwood has high path overlap only at the lower scales of 400–1600m – which may only be due to the effective walking distance from the station – with much less overlap at the larger scales. The segmentation of the centre by the railway line as well as by the major road network seems to have an effect on its grid, with no deformation or intensification around the main roads (Figure 7.7).

In contrast, High Barnet and Surbiton have a much greater grid deformation alongside large roads connecting onwards. In Barnet the live centre is actually on the big road and the circulation is across it, as well as circuitously around it in a form of centrality (as proposed in fact by Hillier as a function of larger, more urban centres; see Hillier 1999a). This is reflected in a strong peak

Figure 7.5
Traffic on South Norwood high street *c.*2008
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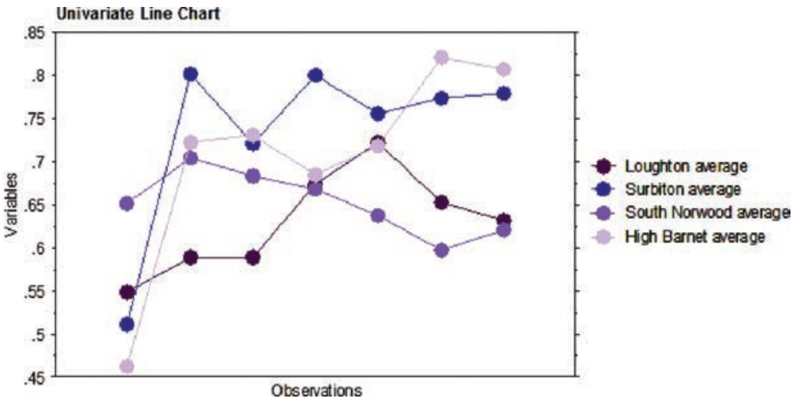

Figure 7.6

Line chart showing overlap between radius 400m, 800m, 1200m, 1600m, 2000m, 3000m and 4000m from left to right

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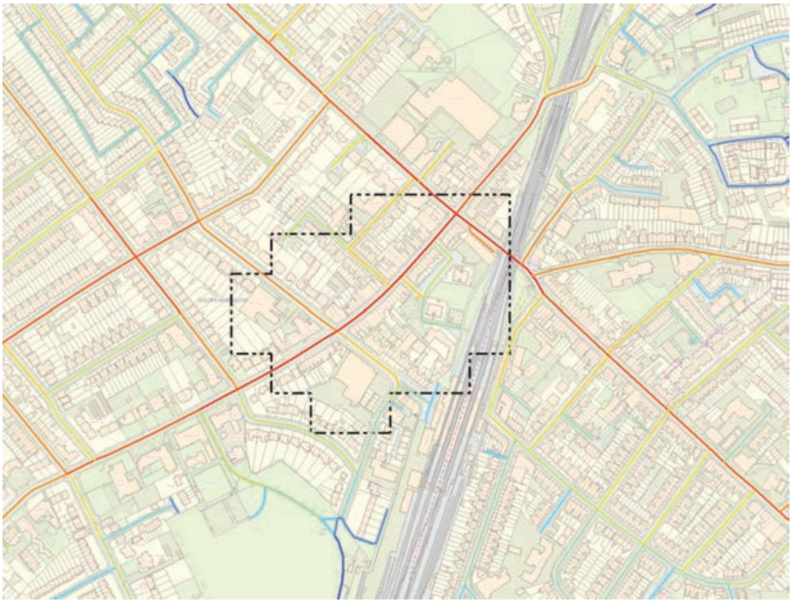

Figure 7.7

Contemporary map of South Norwood street layout surrounding the town centre (with peak activity area marked in black dashes) overlaid with segment angular integration 800m

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of correspondence of scales at the 800 m–1200 m range, but then again at what may be critical: the 3000m–4000m scales. Surbiton has a slightly different pattern on the ground: like South Norwood, it is segmented by the railway line and, at a very large scale, suffers to a certain degree from the Kingston By-Pass; yet possibly because the by-pass links back to historic road alignments, such as Ewell Road, it serves to reconnect what it divides elsewhere – it projects local connections at a higher scale. Like High Barnet it has a strong correspondence at 800 and also at 1600 and then exactly the same at 3000m–4000m.

Putting aside the local differences, which add further veracity to the spatial signature argument mentioned above (Vaughan et al. 2010b), these averages are fair representations of all the different periods. They represent the structural continuity of the built form/movement interfaces in each of these places. By looking at the relationship between the two types of movement pattern, there is supporting evidence for structural continuity across the sample. Arguably this is a definition of structural resilience in terms of the way in which, despite the massive spatial change (consider the doubling and then half again in size of the network, for example – see also Dhanani's chapter in this volume), the fact is that the movement interface index remains consistent over time.

Whilst through-movement is restricted to only the one street in South Norwood, there is a degree of complexity in places such as High Barnet that provides enough local movement, without preventing people from making larger trips across and through it. Having the station situated on the edge rather than cutting through the centre perhaps helps in this instance; what also possibly helps is the fact that the main road runs through the high street without overwhelming it with heavy traffic. There are alternatives, turnings off, that allow for centrality to develop, or at least, the town centre can spread beyond the linear high street. Surbiton is another such case with its three 'high streets' – Brighton Road, Victoria Road and Ewell Road, (notwithstanding Maple Road with its secondary functions) – that allow the town centre to spill over and around the corner in a much more effective way. Loughton has some local centrality either side of its very strong spine of a high street, but despite a large number of pedestrian routes in and around it, it has not extended its growth that far beyond it. Nevertheless, the relatively large amount of non-domestic activity in and around its centre is a sign that, despite its relatively small size, it has sufficient adaptability to weather long-term change. South Norwood is effectively the worst-off of all four cases. It has not managed to adapt its network (or its uses) as well over time, evident in the fact that despite being the closest to the centre of London, and the most urban in its amount of building coverage, it has the least number of centrality features.

V. Building adaptability

The above analysis has shown that the continuity of generic types of activity, despite major changes to the actual functions they maintain, seems to be a consequence of there being a variety of building types, sizes and street morphologies which together are more likely to propagate patterns of co-presence over time – providing the minimal but essential balance between stability of uses, on the one hand, and adaptability in building, as well as use class, on the other. MacCormac's 'reinhabited coral reefs' quoted above are essentially an observation of building scale adaptability, but this phenomenon also occurs at the urban scale. Arguably, neither a rigid structure that only allows (in the extreme case) for one use in a big box, nor a high turnover of businesses which creates an unstable system,

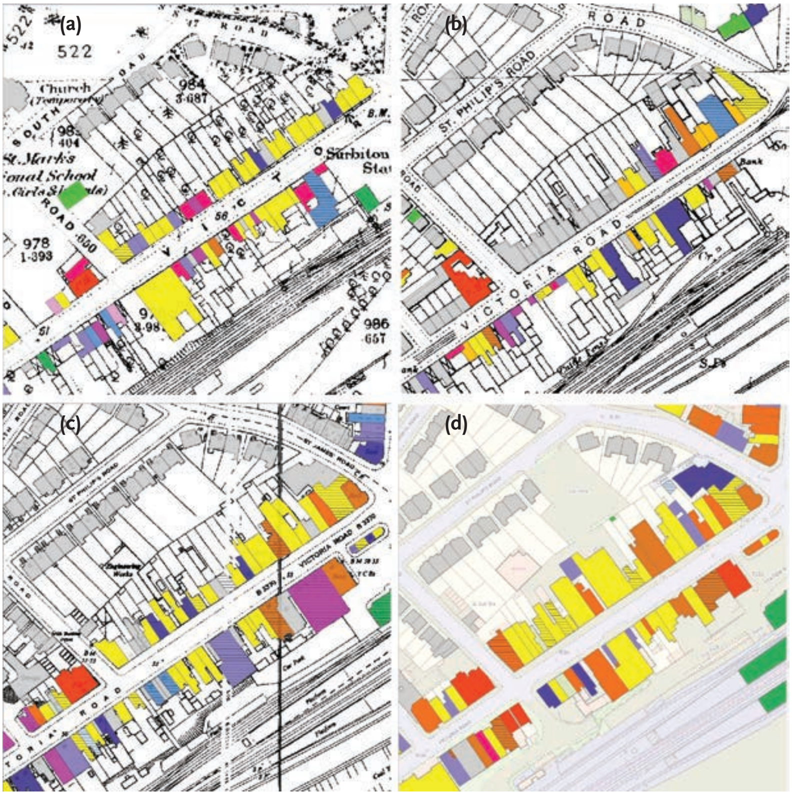

Figure 7.8
Non-domestic land uses, Victoria Road, Surbiton in (a) 1880, (b) 1910, (c) 1960 and (d) 2013. Refer to Fig. 7.4 for land use legend.
Laura Vaughan.

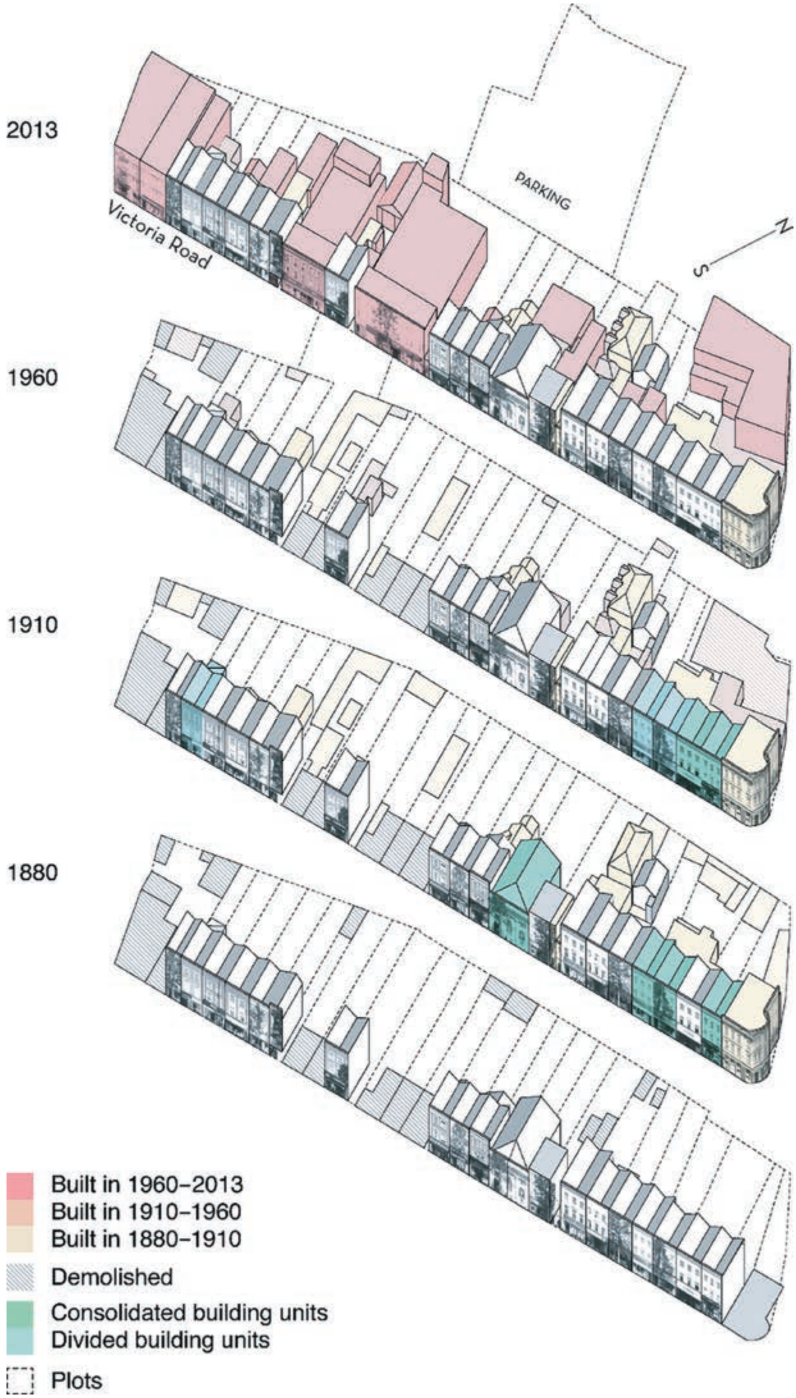

Figure 7.9
Axonometric projection of the built form changes over time on the north side of Victoria Road, Surbiton.

Figure 7.10
(a) Victoria Road, Surbiton, 1907 From the private collection of Dr Ruth Davies and
(b) Victoria Road, Surbiton, 2014 © Ilkka Törmä.

will develop a successful centre. Taking one of the more successful cases of street morphological adaptability outlined above, we can start to explore the built form and land adaptability in a couple of streets in the area.

First, considering a section of Ewell Road in Surbiton, where the highest concentration of changes of uses has taken place over the past 130 years (Figure 7.8), it is evident that the site pre-dates the railway development. Space syntax analysis shows it has one of the highest degrees of 'choice' in every period studied, suggesting it is likely to have benefited from high rates of through-movement throughout the period. Another aspect of its adaptability over time is the building survey undertaken by Ilkka Törmä for the Adaptable Suburbs project, which shows how most of the non-domestic premises are spatially constrained – small extensions to residential buildings, both to the fronts and backs, or small buildings on small plots – but manage to adapt their premises by a variety of means. This can be seen in the axonometric projections in Figure 7.9, which shows the shifting patterns of sub-division into open-plan, multiple-use, rear courtyard uses and partial consolidation, forming together a classic cycle of continuity and change (Whitehand et al. 2014).[4] Such streets are resilient across the scale and Törmä's (2014) study showed how a shorter frontage length, smaller building footprint, smaller plot size and higher plot efficiency typically predict use change. Figure 7.9 shows how the section of street went through the whole gamut of division, consolidation, demolition and extension, but also – alongside a high diversity of activity – a balance between constancy and change in land uses. This illustrates how the persistence of street form is tied up with how buildings and plots are arrayed along the high street. Indeed, if we look at the pair of images in Figure 7.10 of the same street section in 1907 and 2015, other than the greater presence of pedestrians in the earlier period and motor vehicles in the latter, the very persistence of the general ambience of the road is the most striking aspect that one observes.

VI. Conclusion

Biological analogies proliferate in architecture. Perhaps this is because of the non-discursive nature of architectural space. Yet, in this case, the notion of a high street ecology stemmed from a real-life observation of the nature of historical town centres: that these seem to have a richer and more diverse mix of uses and that this diversity seems, on the one hand, to follow a spatial logic and, on the other, to be bound up with the social and economic vitality of these places. Such an ecological analogy to urban systems is effectively restating the proposition that diverse ecosystems are resilient and more able to adapt to change, as detailed in an earlier conference paper regarding this topic (Vaughan et al. 2015).

Bill Hillier has proposed in the past that there is 'a generic process of centre formation' whereby centres start with a spatial seed, such as an intersection or indeed a single road segment which will, by virtue of its spatial configuration, have particular spatial properties that shape its future growth (2009a: 42). This research has provided evidence to support Hillier's proposition that the subsequent growth of centres is dependent on the relative balance of the local and wider-scale ('global' scale) connections of the streets running through them which shapes their degree of interaction with neighbouring and more distant centres. We have seen evidence that the more successful of the four cases studied confirm his theory in the way in which they have evolved over the past 130 years.

The research presented here suggests that a shorter frontage length, smaller building footprint, smaller plot size and higher plot efficiency typically predict use change, but it does not necessarily follow that use change on its own is a measure of resilience, or the adaptability of the buildings to change. Whilst such streets or segments of streets can be active places – like many of the twenty cases studied across our research projects – with continuous non-domestic activity, they are not necessarily resilient in the sense that the same use has been continuous over time. Instead, they are places where uses come and go. One weakness that can be observed is that there are some plots that have become consolidated over time to allow for larger buildings to be erected, which means that in the long term there will be less scope for diversity of uses than in the past. One conjecture that explains this process might be that the limitation in space does not allow for modification to changing needs. Whilst use change takes place at the highest rate in spatially constrained buildings, most distinctly in non-domestic buildings that are on efficiently built-up plots, the analysis also shows that small buildings situated on larger plots are more vulnerable to irreversible change. Overall, what is evident is that the overarching predictor of use change is the accessibility of the street itself as well as the flexibility of the street's area to adapt to changing scales of movement in and around it.

This research also indicates that street network complexity helps contribute to a town centre's resilience against external disruptive forces, such as economic downturns or social change (such as different populations moving into an area), although in comparison to city centres, smaller town centres are likely to be more vulnerable, given their relatively lower connectivity city-wide. In our cases this is characterised by connections via a small number of arterial roads alongside good public transport: over-ground trains and buses. Their lower population densities compared with urban centres means that suburban town centres somehow need to work harder to sustain themselves through economic downturns. At the same time, the long-term growth of cities towards their edges can cause a conflict between older pathways of local movement and an increase of traffic through the area (Jones et al. 2007; Gort Scott Architects and UCL Bartlett School of Planning 2010). Morphological diversity, it is argued, enables the development of niche markets in smaller centres which can support new forms of socio-economic activity.

Notions of diversity cannot be separated from the social properties of urban and suburban built environments. Just as the fundamental intelligibility of the built environment is an essential condition of bringing people together in a public space and affording them encounter, it is a no less, possibly more, vital resource in the relatively sparse environment of the suburban town centre. Having a mix of smaller and larger buildings allows for a mix of businesses of various sizes as well as the array of activities that necessarily feed off each other within a town centre. This is why the study considered high streets within their wider context: land uses within a catchment of up to a kilometre away and built form and network connectivity within a radius of 3km. By taking account of the larger spatial ecology, we were able to explore the full extent of the interconnected relationships between land uses and the people who serve and use them.

Evidently diversity is not the same as 'mixed-use'. The spatial structure of the pattern of inter-relationships, from building to street, from street to neighbouring street and from local streets to the wider network, works to create an interdependence of activity much wider than that observed by MacCormac. The significance of this finding is greater still if we consider that the discussion of diversity has mainly focused here on land use diversity. We might legitimately propose and use diversity as a proxy for co-presence. It is easy to overlook this self-evident aspect of land use diversity: the greater the diversity of land uses, the more likely it is to generate different sorts of activity by different sorts of people.

If we consider for a moment social diversity, folklore has it that Paris organised social classes vertically in urban courtyards, from the bourgeois on the main floor to the poor in the upper garret (Bernard 2014).[5] Berlin did it horizontally, with the rich at the front and the poor at the back, and London arranged classes turn-by-turn: linearly integrated, but marginally separated, as can be seen in the famous Booth map of poverty, where different street alignments each carried different poverty classes. As Hillier (1996: 166) has suggested, all of these are formulae for mixing uses without them getting in each other's way. In the case of London, *street alignment*, not the *block*, has tended to be the basic organising urban element. Different uses or grades of use are juxtaposed but on different streets, separated just enough to allow for the different grid conditions to play their part in generating higher amounts of movement on the main streets (where more public-facing activities can be located) and lower amounts of movement on the back streets. The process of the formation of these spatial patterns, even on the outskirts of the city, is evidently key to the continuity and robustness of London's town centres, as has been shown in this chapter.[6]

Notes

1 Some of the research contained in this chapter was presented as a paper at the 10th International Space Syntax Symposium in July 2015 (Vaughan et al. 2015).

2 Interestingly, Penn et al.'s research on land use patterns in Clerkenwell points to the possibility that diversity could sometimes emerge as a result of economically marginal locations, allowing for places 'just off' the main throng to emerge as loci of specialist businesses. Historical analysis of long-standing market places such as Exmouth Market in Clerkenwell support this contention, showing that Exmouth Market used to be the main thoroughfare of the area and it is only when a new street alignment was inserted to its north-west to carry trams that the market street came to be cut off at one end and effectively receded into the background.

3 Kelly's directory (1901) lists the following, from the Surbiton Hill junction, southwards: wine & spirit merchants, physicians & surgeons, dentist, surgeon, … physicians & surgeons, Young Women's Christian Association, builder, decorator, cycle maker, oilman, saddler, stationer, … linen draper, fruiterer, corn dealers, grocer, school, fancy repository, provision merchants, chemist, grocers & provision merchants, Surbiton Hill Post & Telegraph Office, baker, butcher, dairyman, fishmonger, dairyman, boot & shoe warehouse, grocers & provision merchants, … girls' school, … Urban District Council of Surbiton, Wesleyan Chapel, solicitor, veterinary surgeon, School of Art, Anchor Coffee Tavern, butchers, fly proprietor, Plough Public House, ironmonger, boot maker, milliner, beer retailer, plumber, builders, upholsterer, umbrella maker & cutler, cycle company, newsagent, butcher, … Victoria Public House, draper, coal merchant, London & Provincial Bank Ltd., builders, South Down Farm Dairy … etc.

4 The urban morphologist Jeremy Whitehand outlines two kinds of urban landscape change. 'One is the adaptation (in a broad sense) of existing forms: this includes, for example, the modification of a building, perhaps by adding rooms or changing the roof, or, in the case of a street, it might involve changing the relationship between the areas allocated to vehicles and those for pedestrian use only. Changes of this type may occur at various intervals over the lifespan of a plot, building, street, or other feature in the landscape. An example well known among geographical urban morphologists is the burgage cycle. In this cycle the "burgages" – the plots of land derived from those held by the enfranchised members of medieval boroughs – become progressively built up. Each plot has a building constructed at its head, usually early in the life of the plot, and this is followed by subsidiary buildings being added along the length of the plot over many decades, or sometimes over centuries …
The other kind of change is the introduction of new types of buildings, streets or other features. In England, bye-law terraced houses introduced and reproduced in the Victorian and Edwardian periods, followed by semi-detached houses (of "universal" plan) in the interwar period, are well-known "period types", at least in that part of the world. The tower block and the expressway (to use the American term) were widespread introductions, essentially in the twentieth century, although introduced in different parts of the world at different times.'

5 Andreas Bernard (2014) argues that whilst in the nineteenth century the attic and basements of buildings attracted a stigma of abnormality, by the early twentieth century, the invention of the lift in fact 'freed the upper storeys [of hotels] from the stigma of inaccessibility and lent them an unheard-of glamour', meaning that the highest rental prices could be charged for penthouses.

6 Indeed, both clustering and separation can co-exist, as shown in historical research into immigrant settlement patterns (Gilliland et al. 2011: 493, 495).